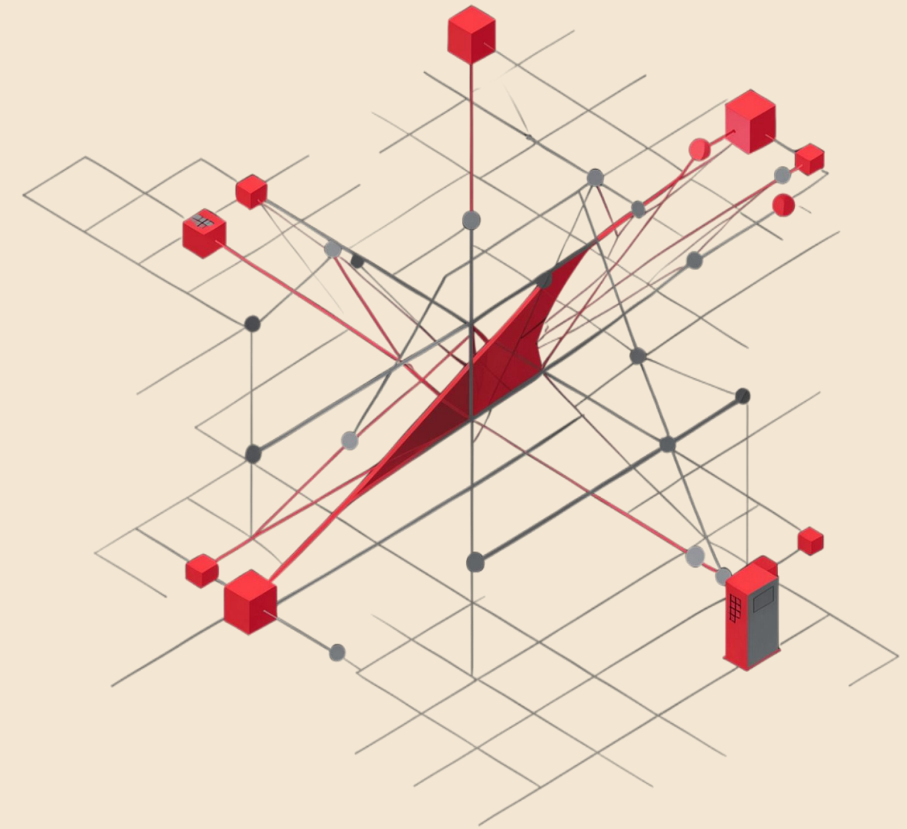
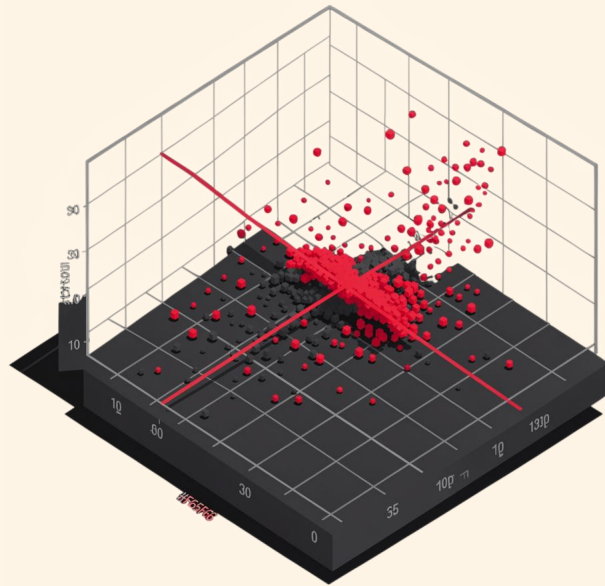


Support Vector Machine (SVM) in Machine Learning

A Powerful Supervised Learning Algorithm for Classification and Regression





What is SVM?

→ Definition

A supervised learning algorithm that finds the **optimal decision boundary (hyperplane)** to separate classes

→ Why "Support Vector"?

The boundary is defined by the **closest data points** — called *support vectors* — which "support" the hyperplane

→ Tasks & Uses

Classification (SVC) and Regression (SVR) — applied in **face recognition, spam detection, cancer diagnosis**, and image classification

How SVM Works — Step by Step



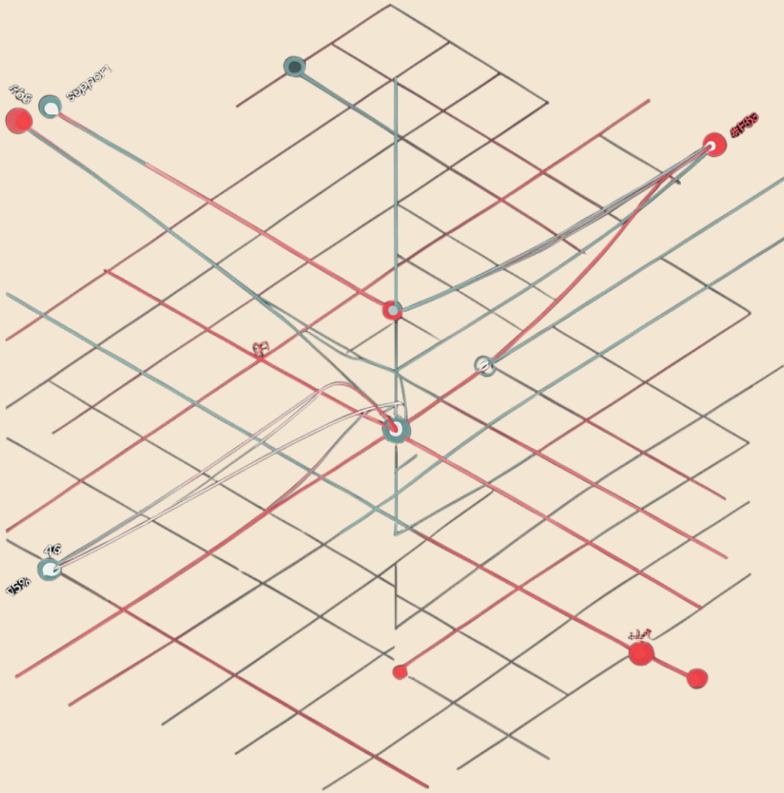
Training Data

Feature Space

Support Vectors

Hyperplane

SVM iteratively refines the decision boundary by identifying support vectors and solving an optimization problem to maximize the separation margin between classes.



Key Concepts of SVM

Hyperplane

Decision boundary separating classes — a line in 2D, a plane in 3D, a hyperplane in n-D

Support Vectors

Data points closest to the hyperplane — removing them changes the boundary entirely

Margin

Distance between the hyperplane and the nearest support vectors on each side

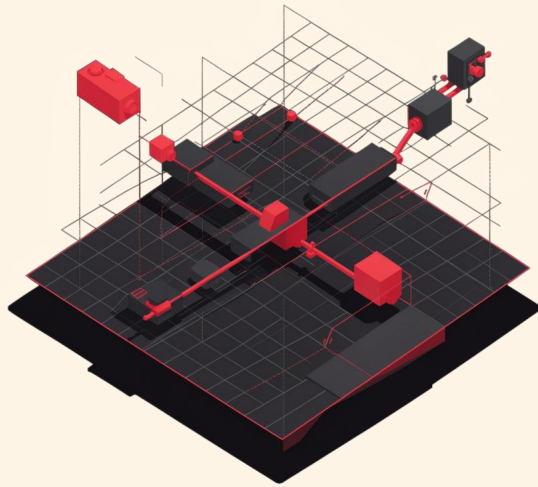
Maximum Margin

SVM maximizes this margin → better generalization and less overfitting

Kernel Functions — Handling Non-Linear Data

The Kernel Trick

Real data is rarely linearly separable in its original space. The **kernel trick** maps data into a higher-dimensional space where a linear boundary *can* separate classes — without computing the transformation explicitly.



Kernel	Formula (intuition)	Best For
Linear	$x \cdot x'$	Linearly separable data
Polynomial	$(\gamma x \cdot x' + r)^d$	Curved boundaries, NLP
RBF / Gaussian	$\exp(-\gamma \ x - x'\ ^2)$	Non-linear, most common
Sigmoid	$\tanh(\gamma x \cdot x' + r)$	Neural network-like tasks

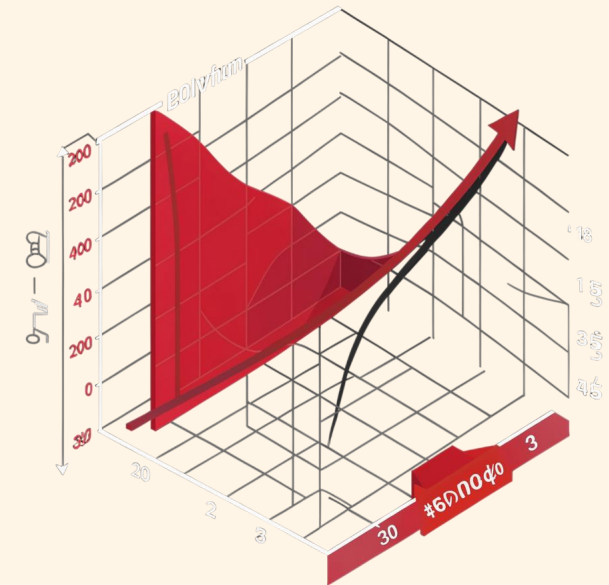
Model Evaluation Metrics

Confusion Matrix

	Predicted Positive	Predicted Negative
Actual Positive	TP (True Positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

Key Metrics

- **SVC:** Accuracy, Precision, Recall, F1 Score, AUC
- **SVR:** MAE, RMSE, R^2 Score
- **k-Fold CV** prevents overfitting on unseen data



Advantages & Limitations

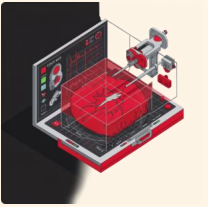
✓ Advantages

- **High Accuracy**
Effective in high-dimensional feature spaces
- **Small Datasets**
Works well even with limited training data
- **Robust**
Resistant to overfitting via margin maximization

✗ Limitations

- **Computationally Expensive**
Slow to train on very large datasets
- **Kernel Selection**
Choosing the right kernel requires domain expertise
- **Less Interpretable**
Harder to explain than decision trees or logistic regression

Real-World Applications of SVM



Healthcare

Cancer detection, disease diagnosis from medical imaging



Finance

Credit risk scoring, fraud detection in transactions



Computer Vision

Face recognition, image classification



Cybersecurity

Spam filtering, network intrusion detection



NLP

Sentiment analysis, document classification



Bioinformatics

Gene expression classification, protein structure analysis

SVM vs K-Nearest Neighbors (KNN)

Criterion	SVM	KNN
Learning Type	Parametric (model-based)	Non-parametric (instance-based)
Training Time	Moderate-High	Very Fast (lazy learner)
Prediction Time	Fast	Slow (searches all points)
Accuracy	High (with right kernel)	Moderate
Memory Usage	Low (stores support vectors only)	High (stores all data)
Scalability	Moderate	Poor on large datasets
Decision Boundary	Hyperplane (global)	Local neighborhood
Interpretability	Low	High
Best Use Case	High-dim, small-medium data	Low-dim, small datasets

Summary & Key Takeaways

SVM is a **supervised learning algorithm** for classification and regression

Finds the **optimal hyperplane** that maximizes the margin between classes

Kernel functions extend SVM to non-linear problems via the kernel trick

Excels in **high-dimensional spaces** with small-to-medium datasets

Applied across **healthcare, finance, cybersecurity, vision, and NLP**

"Support Vector Machines classify data by finding the optimal boundary that maximizes separation between classes — making them one of the most accurate and reliable machine learning algorithms."

Thank You — Questions?